ARISTOTLE UNIVERSITY OF THESSALONIKI

FACULTY OF ENGINEERING – SCHOOL OF MECHANICAL ENGINEERING

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SEMESTER: 8th

COURSE: DATA ANALYSIS

2nd ASSIGNMENT

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Note: tables and figures are referenced by their original labels (e.g. "Table 1", "Image 1"). Replace each `\figplaceholder` with your own output via `\includegraphics`.

1. Introduction

The present assignment constitutes the second part of the data analysis concerning wine production from the file `Data_Analysis_2026.csv`. In it, the relationships between the predictor variables (predictors) and the response variable, which is the quality of the wine, will be investigated. The remaining variables will be used as predictors so that conclusions can be drawn about whether and to what extent they affect quality. As can be understood, this part of the assignment focuses on regression and its branches (simple, multiple linear, non-linear).

2. Question 1

The first question asks for the investigation of each variable (predictor) separately in simple linear regression models, and whether it is statistically significant, always in relation to the response variable, quality.

The questions that will be answered in this particular question are:

- Whether there is a relationship, and how strong it is, between the predictor and the quality,
- Which predictors increase the quality,
- How accurate the estimate of the effect is,
- Is the relationship linear?

It is obvious that the starting point of the problem is whether there is a relationship between the predictor and the response variable, so that the subsequent computational and diagrammatic investigation makes sense. First, β_0 (constant) and β_1 (coefficient) will be computed for each relationship so that its formula is completed.

$$y = \beta_0 + \beta_1 x$$

The coefficient β_1 expresses the change in wine quality for a one-unit increase in the independent variable. The final judge of the analysis, as always in statistics, will be the p-value of the correlation, considering relationships with p-value < 0.05 as statistically significant. The smaller it is, the stronger the indication of correlation. But to arrive at this, we will need to compute: the coefficient (regression coefficient β_1), the Standard Error, and the t-statistic (coefficient / Std. Error).

[Table 1 — FINAL TABLE: per-variable statistics (coefficient, std error, t, p-value)]

Table 1

In Table 1, the statistical analysis is presented, and it is observed that all predictors, except chlorides, pH and sulphates, have statistically significant indications of dependence of quality — some to a large degree (alcohol 6.26e-253) and others to a lesser degree (1.82e-2). Their diagrams are also presented in the appendix. Table 2, meanwhile, gives the formula of the simple linear regression for each variable.

[Table 2 — simple linear regression model per variable]

Table 2

3. Question 2

To investigate the relationship between the response variable (quality) and all the candidate predictor variables, a multiple linear regression model of the following form was estimated:

$$\text{quality} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$$

where X_i are the predictor variables and β_i the corresponding coefficients.

The model was estimated using the ordinary least squares (OLS) method. The significance of each coefficient was checked through the p-value. For p-value < 0.05, the null hypothesis $H_0: \beta_j = 0$ is rejected, which indicates that the corresponding variable affects the wine quality in a statistically significant way. From the results it follows that the variables alcohol, volatile acidity, density, fixed acidity, free sulfur dioxide, pH, residual sugar, sulphates and total sulfur dioxide are statistically significant ($p < 0.05$), therefore the null hypothesis is rejected for these. On the contrary, the variables chlorides and citric acid are not statistically significant, since their p-values are greater than 0.05, and therefore H_0 is not rejected for them. The coefficients of the multiple regression (coefficient b_1) express the effect of each variable on the wine quality, keeping the remaining variables constant. For example, the variable alcohol has a positive effect on quality, while volatile acidity has a negative effect.

[Image 2.1 — insert your figure here]

Image 2.1

For the evaluation of the model, the coefficient of determination R^2 was used, which shows the percentage of the variability of quality explained by the model. The value of R^2 (0.287) indicates a moderate fit, which means that the wine quality is not fully explained by the examined variables. In addition, the Adjusted R^2 (0.285) confirms the reliability of the model, taking into account the number of variables. Finally, the F-statistic was used for the overall significance test of the model.

[Table 3 — OLS Regression Results (full summary)]

Table 3

[Table 4 — significant variables list]

Table 4

4. Question 3

For the selection of the optimal model, the forward selection method was applied. The process started from the null model, and at each step the variable that led to the greatest improvement of the model was added, based on the smallest value of the RSS. The process continued until no further substantial improvement resulted and the new variable was no longer statistically significant. According to the final model, the selected variables were: alcohol, volatile acidity, sulphates, residual sugar, total sulfur dioxide, free sulfur dioxide, density, pH and fixed acidity, and the final coefficient of determination was $R^2 = 0.2865$, with the Adjusted R^2 differing by 0.001. This is reasonable, since only statistically significant variables are used and the rest are ignored. In the end, 9 of the 11 variables were included, since enriching the model with the remaining 2 did not offer any substantial improvement. The two variables that were rejected are chlorides and citric acid, which had excessively high p-values, much greater than the 0.05 limit.

[Table 5 — forward selection steps (added variables, RSS, p-value)]

Table 5

[Table 6 — selected variables, final R^2 / Adjusted R^2 / RSS]

Table 6

5. Question 4

To investigate a non-linear relationship between quality and Residual sugar, chlorides and alcohol, linear, quadratic (X^2) and cubic (X^3) regression models will be estimated successively. The selection of the most appropriate model was based on the statistical significance of the (X^2, X^3) terms and on the corrected coefficient Adjusted R^2. Because X, X^2, X^3 are highly correlated with each other, there may be unstable coefficients or unexpected p-values due to collinearity.

From the results it follows that, for all three variables, the most appropriate model is the cubic (X^3), since it presents the highest Adj R^2 index.

Specifically, the variable residual sugar shows low Adjusted R^2 indices for the three models (0.001280, 0.001597, 0.007999 for the linear, quadratic and cubic models respectively), of which the cubic model appears the "strongest". Nevertheless, the index is excessively low and cannot be taken into account as an indication of a non-linear relationship.

In the case of the variable chlorides, the linear model appears to be considerably weaker than the quadratic and the cubic, by one order of magnitude, since the values they take are -0.000044, 0.043679, 0.062658. The linear model has a zero or even negative adj R^2 value, which indicates its weakness and unsuitability. On the contrary, the non-linear models "fit" the data better and reveal a non-linear relationship between chlorides and the prediction of quality.

Finally, for the variable alcohol, the R^2 and Adjusted R^2 indices are quite high for all three models, exceeding the value 0.20 in each model, which reveals the strong correlation of the variable with the response value, quality. Nevertheless, the linear appears to be the weakest (0.204460), followed by the quadratic (0.204642), and the cubic appears the strongest (0.211219). The increase of the index is not justified by the increase in the complexity of the model, and based on the principle of simplicity (parsimony), the non-linear models are rejected, since the linear one satisfactorily explains their correlation.

[Table 7 — summary table (R^2 and Adj R^2 for linear / quadratic / cubic per variable)]

Table 7

6. Question 5

The use of linear regression constitutes a suitable initial approach for studying the relationship between wine quality and the remaining variables given as data, since it allows the estimation of coefficients, the testing of statistical significance, and the interpretation of the effect of each variable. The results showed that certain factors, such as alcohol, have a strong and statistically significant relationship with quality, which confirms the usefulness of the model. Nevertheless, limitations were identified, such as indications of non-linearity in certain variables, possible multicollinearity, and relatively low R^2 values, which suggest that quality is not fully explained by the examined variables. In addition, the existence of different wine types may affect the results. Therefore, the regression model is useful as an analysis tool, but requires careful interpretation and possible extensions.

Appendix

Simple Linear Regression Diagrams

The simple linear regression scatter plots (quality vs each predictor, with fitted line) referenced in Question 1 are collected here, matching the original numbering (Images 1–8).

[Image 1 — Quality vs citric acid]

Image 1

[Image 2 — Quality vs density]

Image 2

[Image 3 — Quality vs alcohol]

Image 3

[Image 4 — Quality vs fixed acidity]

Image 4

[Image 5 — Quality vs free sulfur dioxide]

Image 5

[Image 6 — Quality vs residual sugar]

Image 6

[Image 7 — Quality vs total sulfur dioxide]

Image 7

[Image 8 — Quality vs volatile acidity]

Image 8